3.13 Human Immunodeficiency Virus (HIV) Related Opportunistic Infections

ADULTS

CHAPTER 3.13	HUMAN IMMUNODEFICIENCY VIRUS-RELATED OPPORTUNISTIC INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Preferred	**Alternative**	
***Pneumocystis Jirovecii* (*Carinii*) Pneumonia (PJP/PCP)**			
Treatment	**Moderate to Severe Disease** Co-trimoxazole (Trimethoprim component) 15 - 20 mg/kg/day IV divided q6 - 8H for 21 days May switch to PO after clinical improvement	**Moderate to Severe Disease** Clindamycin 600 mg IV q6H **OR** 900 mg IV q8H + Primaquine base 30 mg PO q24H for 21 days **OR** Pentamidine 4 mg/kg IV q24H infused over > 60	• Duration: 21 days • Indication for steroids: PaO_2 < 70 mm Hg (9.3kPa) or SpO_2 < 92% on room air. • Prednisolone doses (beginning as early as possible and within 72 hours of PCP therapy): **Days 1 - 5:** 40 mg PO q12H **Days 6 - 10:** 40 mg PO q24H **Days 11 - 21:** 20 mg PO q24H • If unable to take oral, methylprednisolone at 75% of this dose can be administered. • Patients given dapsone should be tested for G6PD deficiency.

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		mins	• Co-trimoxazole and clindamycin has excellent oral bioavailability which may be switched to PO after clinical improvement.
	Mild to Moderate Disease Co-trimoxazole (Trimethoprim component) 15 - 20 mg/kg/day PO divided q8H for 21 days	**Mild to Moderate Disease** Clindamycin 450 mg PO q6H **OR** 600 mg PO q8H + Primaquine 30 mg base PO q24H **OR** **If unable to tolerate to above regimens with risk of kidney impairment and/or cytopenia:** Consider lower dose of Co-trimoxazole (Trimethoprim component) 7.5 - 10 mg/kg/day IV divided q6 - 8H for 21 days, may	

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		switch to PO after clinical improvement.	
Secondary prophylaxis	Co-trimoxazole (480 mg) 1 - 2 tablets PO q24H	Dapsone 100 mg PO q24H **OR** Pentamidine 300 mg nebulised every month	• **Discontinuation of secondary/ primary prophylaxis** • CD4 count increased from <200 cells/mm^3 to >200 cells/mm^3 for >3 months as a result of ART **OR** • Can consider if CD4 count is 100–200 cells/mm^3 and HIV RNA remains below limits of detection of assay used for ≥3 months to 6 months.
***Toxoplasma gondii* encephalitis**			
Treatment	Pyrimethamine 200 mg PO single dose **Followed by** Pyrimethamine 50 mg (< 60 kg) **OR** 75 mg (≥ 60 kg) PO q24H +	**Sulfonamide Allergy:** Pyrimethamine 200 mg PO single dose **Followed by** Pyrimethamine 50 mg (< 60 kg) **OR** 75 mg (≥ 60 kg) PO q24H	• Duration: for at least 6 weeks. • Longer duration if clinical or radiological disease is extensive or response is incomplete at 6 weeks. • Adjunctive corticosteroids (e.g., Dexamethasone) should be administered when clinically indicated, e.g., for treatment of a mass effect associated with focal lesions or associated oedema. Discontinue as soon as clinically feasible.

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	Sulfadiazine 1 g (< 60 kg) **OR** 1.5 g (≥ 60 kg) PO q6H + Folinic Acid 15 mg PO q24H **OR** Co-trimoxazole* (Trimethoprim component) 5 mg/kg IV/PO q12H	+ Folinic Acid 15 mg PO q24H + Clindamycin** 600 mg IV/PO q6H	• *Use Co-trimoxazole only when Pyrimethamine is not available. • **If Clindamycin is used in place of Sulfadiazine, additional therapy must be added to prevent PCP.
Secondary prophylaxis	Pyrimethamine 25 – 50 mg PO q24H + Sulfadiazine 0.5 - 1 g PO q6H + Folinic Acid 15 mg PO q24H	Pyrimethamine 25 – 50 mg PO q24H + Clindamycin 600 mg PO q8H + Folinic Acid 15 mg PO q24H	• **Discontinuation of secondary prophylaxis:** • Consider when on ART, CD4 > 200 for more than 6 months and viral load suppressed.

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	OR Co-trimoxazole (480 mg) 2 tablets PO q12H		
Cytomegalovirus (CMV) Infections			
Cytomegalovirus (CMV) retinitis			
Initial therapy	Ganciclovir 5 mg/kg IV q12H for 2-3 weeks **OR** Valganciclovir 900 mg PO q12H for 2-3 weeks **Immediate Sight Threatening Lesions** Intravitreal Ganciclovir may be administered with consideration of placement of intravenous Ganciclovir	Foscarnet 90 mg/kg IV q12H for 2-3 weeks*	• *Foscarnet is not available in UMMC formulary. • Initial therapy should also include optimisation of ART.

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	Intraocular implant provides more expedient loading doses and in patients unable to tolerate systemic therapy.		
Chronic maintenance therapy	Ganciclovir 6 mg/kg IV q24H 5 times weekly **OR** Valganciclovir 900 mg PO q24H	Foscarnet 120 mg/kg IV q24H 5 times weekly*	• *Foscarnet is not available in UMMC formulary. • Maintenance therapy for CMV retinitis can be safely discontinued in patients with inactive disease and sustained CD4+ count (>100 cells/mm^3 for ≥ 3 – 6 months) and undetectable viral loads. • Patients with CMV retinitis who discontinued maintenance therapy should undergo regular eye examination, optimally every 3 months, for early detection of relapse or immune recovery uveitis (IRU).
Cytomegalovirus (CMV) oesophagitis or colitis			
Treatment	Ganciclovir 5 mg/kg IV q12H	Foscarnet 90 mg/kg IV q12H*	• Duration: 21 to 42 days or until resolution of signs and symptoms.

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	OR Valganciclovir 900 mg PO q12H		• Maintenance therapy is usually not necessary. • ART offers best hope for prevention of relapses. • *Foscarnet is not available in UMMC formulary.
Cytomegalovirus (CMV) neurological disease			
Treatment	Ganciclovir 5 mg/kg IV q12H + Foscarnet* 90 mg/kg IV q12H **OR** 60 mg/kg IV q8H	Valganciclovir 900 mg PO q12H	• Optimal duration of therapy has not been established. It is recommended to continue until CMV PCR has become undetectable in both CSF and serum and clinical evidence of disease has resolved. • *Foscarnet is not available in UMMC formulary.
Cryptococcal meningitis *Cryptococcus neoformans var neoformans*			
Induction Therapy	**Regimen 1:** Amphotericin B Deoxycholate 1 mg/kg IV q24H	**If renal impairment on therapy or high likelihood of renal failure:** Liposomal Amphotericin	• *To consider Amphotericin B Lipid Complex or **daily** Liposomal Amphotericin B in those with renal dysfunction on therapy or have high likelihood of renal failure. • **The single, high-dose regimen of Liposomal Amphotericin

Infections and Likely Organisms	Suggested Treatment		Notes
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	OR Amphotericin B Lipid Complex 5 mg/kg IV q24H* + Flucytosine 25 mg/kg/dose PO q6H (total 100 mg/kg/day) x 1 week **Followed by,** Fluconazole PO 1200 mg q24H x 1 week **Regimen 2:** Liposomal Amphotericin B 10 mg/kg IV single dose on D1** **Followed by,** Flucytosine 25 mg/kg/dose PO q6H (total 100 mg/kg/day) x 2 weeks	B 3-4 mg/kg IV q24H* + Flucytosine 25 mg/kg/dose PO q6H (total 100 mg/kg/day) x 1 week **Followed by,** Fluconazole PO 1200 mg q24H x 1 week **If Amphotericin B cannot be used/not available:** Fluconazole 1200 mg IV/PO q24H + Flucytosine 25 mg/kg PO q6H for 2 weeks **If Flucytosine not**	B has only been clinically proven for HIV-infected individuals. • Addition of Flucytosine to Amphotericin B has been associated with more rapid sterilisation of CSF, decreased risk for subsequent relapse and reduced mortality. • Opening pressure (OP) should always be measured when a lumbar puncture (LP) is performed. If OP > 250 mm H2O and signs of cerebral oedema present, do daily LPs to reduce CSF pressure. • If clinical signs of cerebral oedema do not improve after about 2 weeks of daily LPs, consider placement of a lumbar drain or ventriculoperitoneal shunt.

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	+ Fluconazole 1200 mg IV/PO q24H x 2 weeks	**available:** Amphotericin B Deoxycholate 1 mg/kg IV q24H + Fluconazole 1200 mg PO q24H for 2 weeks	
Consolidation therapy After at least 2 weeks of successful induction – defined as significant clinical improvement & negative CSF culture.	Fluconazole 800 mg PO q24H for 8 weeks		
Secondary prophylaxis	Fluconazole 200 mg PO q24H until CD4 count > 200 with undetectable viral loads	**Intolerant or failed Fluconazole:** Itraconazole 200 mg PO q12H until CD4 > 200	• **Criteria for Discontinuation of secondary prophylaxis:** • At least 1 year from initiation of antifungal therapy • Patient remains asymptomatic from cryptococcal infection • CD4 count ≥ 100 cells/mm3 and suppressed HIV RNA in response to effective ART.

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***Mycobacterium avium* complex (MAC)**			
Treatment	**At least 2 drugs as initial therapy with:** Azithromycin 500 mg PO q24H + Ethambutol 15 mg/kg PO q24H **Addition of rifabutin or rifampicin may also be considered:** Rifabutin 300 mg PO q24H **OR** Rifampicin 10 mg/kg PO q24H	**At least 2 drugs as initial therapy with:** Clarithromycin 500 mg PO q12H + Ethambutol 15 mg/kg PO q24H **Addition of a 3rd or 4th drug should be considered for patients with disseminated disease advanced immunosuppression (CD4 count < 50) or in the absence of effective ART:** Amikacin 10 -15 mg/kg IV q24H **OR**	• **Discontinuation:** • At least 12 months of therapy, can discontinue if no signs and symptoms of MAC disease and sustained (>6 months) CD4 count >100 cells/mm^3 in response to ART.

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		Streptomycin 15 mg/kg (max 1 gm) IV / IM q24H **OR** Levofloxacin 500 mg PO q24H **OR** Moxifloxacin 400 mg PO q24H	
Talaromycosis (formerly Penicilliosis) *Talaromyces marneffei*			
Induction phase	Amphotericin B Deoxycholate 0.6-0.7 mg/kg IV q24H for 2 weeks **OR** Amphotericin B Lipid Complex* 5 mg/kg IV q24H	Voriconazole 6 mg/kg IV q12H for 1 day then, 4 mg/kg IV/PO q12H for 2 weeks	• Consider discontinuation among patients with CD4 count > 100 cells/mm3 for > 6 months. • Itraconazole syrup has better bioavailability compared to Itraconazole capsule. • *Consider in those with renal dysfunction on therapy or have high likelihood of renal failure
Continuation phase	Itraconazole 200 mg PO		

Infections and Likely Organisms	Suggested Treatment		Notes
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	q12H for 10 weeks		
Chronic maintenance therapy	Itraconazole 200 mg PO q24H		
Candidiasis (Mucocutaneous)			
Oropharyngeal candidiasis, initial episode	**Oral Therapy:** Fluconazole 100 mg PO q24H **Topical Therapy:** Nystatin suspension 100,000 unit/mL, 4–6 mL q6H **OR** 1–2 flavoured pastilles 4–5 times daily*	Itraconazole 200 mg PO q24H	• Duration: 7-14 days • *Nystatin flavoured pastilles is not available in UMMC formulary.
Oesophageal candidiasis	Fluconazole 100 mg (up to 400 mg) IV/PO q24H		• Duration: 14-21 days
Uncomplicated vulvo-	Fluconazole 150 mg PO		

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vaginal candidiasis	single dose **OR** Topical azoles (Clotrimazole or Miconazole) for 3–7 days		
Tuberculosis *Mycobacterium tuberculosis*			
Empirical treatment should be initiated and continued in whom TB is suspected until all diagnostic work up is complete.	**Treatment of drug susceptible active TB disease:** **Initial Phase (2 months)** INH + RIF + PZA + EMB **Continuation Phase** INH + RIF daily **Fixed Dose Combinations (FDC) available:**	**Treatment of drug-resistant active TB:** **Resistant to INH** RIF + EMB + PZA + Levofloxacin 750 mg PO q24H for 2 months **Followed by,** RIF + EMB + Levofloxacin 750 mg PO q24H for 4 months **Multidrug Resistant (MDR) including RIF**	• All HIV-TB co-infected patients should receive co-trimoxazole preventive therapy for the duration of TB treatment. • Adjuvant corticosteroids should be added when treating CNS and pericardial disease. • **Recommended corticosteroid regimens:** • Dexamethasone 0.3 - 0.4 mg/kg tapered over 6 - 8 weeks • Prednisolone 1 mg/kg for 3 weeks then tapered over 3 - 5 weeks • Daily dosing of anti-TB treatment is preferred for HIV infected patients in intensive phase. • Intermittent dosing in intensive phase: 5 days per week for

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	Akurit-4 contains: RIF 150 mg, INH 75 mg, PZA 400 mg, EMB 275 mg **Akurit-2 contains:** RIF 150 mg, INH 75 mg **PLUS** Pyridoxine 10-50 mg PO q24H	**resistant TB or Extensively Drug Resistant (XDR) TB** Therapy should be individualised based on resistance pattern and with close consultation with a respiratory specialist	40 doses by DOT may be considered. • Rifampicin is not recommended in patients receiving protease inhibitors (PI) because of induction of PI metabolism. • Rifabutin is preferred in patients receiving PIs as it is a less potent inducer of CYP3A4. • Rifabutin dose when given with PIs: 150 mg PO q24H. • **Duration of therapy:** • Pulmonary TB: 6 months. • Extrapulmonary TB with CNS, bone, or joint infections: 9 - 12 months. • Extrapulmonary TB in other sites: 6 - 9 months.

Antituberculous drugs dosing:

Drug	Short Form	Daily Dose (Range) mg/kg	Maximum Daily Dose (mg)
Isoniazid	INH	5 (4-6)	300
Rifampicin	RIF	10 (8-12)	600
Pyrazinamide	PZA	25 (20-30)	2000

Ethambutol	EMB	15 (15-20)	1600
Streptomycin	STR	15 (12-18)	1000

Fixed-dose combinations (FDC) dosing:

Weight (kg)	Dose
30-37	2 tabs daily
38-54	3 tabs daily
55-70	4 tabs daily
>70	5 tabs daily